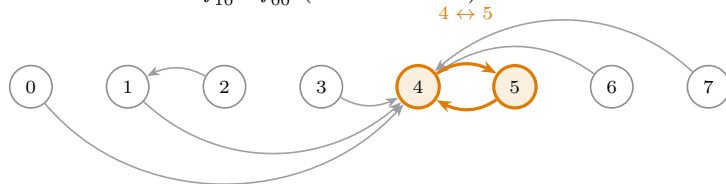


a reading a state label

w	$x_1 x_2 y$	circuit reading
0	000	quiescent
1	001	gate only
2	010	E_2 wins, gate silent
3	011	E_2 wins, gate engaged
4	100	E_1 wins, gate silent
5	101	E_1 wins, gate engaged
6	110	both active, gate silent
7	111	both active, gate engaged

b WTA alone: $h = f_{10}^W \circ f_{00}^W$ (word $W00\ W10$)



every state falls into the pair $\{4, 5\}$: E_1 has won, and h toggles only the inhibitory gate ($4 \leftrightarrow 5$). On the image, h is the transposition $(4\ 5)$: reversible, though neither f_{00}^W nor f_{10}^W is. the symmetric word $W00\ W01$ gives the E_2 cycle $\{2, 3\}$.

WTA state $w = 4x_1 + 2x_2 + y$; composite state $(D:d, W:w)$ has index $8d + w$.

c WTA $\rightarrow$ DN cascade: $t = G_1 \circ G_0$ (word $g_0\ g_1$) on the product space

